

Jaden (Minjae) Moon

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Research interests: Reliable ML; selective prediction; uncertainty quantification; distribution shift; multimodal health sensing.

Education

Dartmouth College, B.A. in Computer Science — Hanover, NH

Expected June 2027

GPA: 3.92 / 4.00

Relevant Coursework: Deep Learning; Deep Learning Robustness; Machine Learning; AI Agents; Software Design & Implementation; Linear Algebra; Differential Equations

Publications

J. Moon, A. Pillai, and A. T. Campbell. (2026). *When Does Quality-Aware Multimodal Fusion Matter? A Leakage-Safe Diagnostic for Decision-Level Dependence*. *INTERSPEECH 2026*. [arXiv] **Accepted**

J. Moon, Y. Wu, A. Pillai, and A. T. Campbell. (2026). *ReliaGate: Reliability Routing for Low-Stakes Wearable Stress Prediction*. *WellComp 2026 (UbiComp/ISWC 2026 Workshop)*. **Accepted**

Selected Research Experience

HealthX Lab, Dartmouth College

Sep 2025 – Present

- **ReliaGate (accepted at WellComp 2026):** Developed a post-hoc reliability-routing framework combining confidence, signal quality and modality trust, cross-modal agreement, train-standardized atypicality, and train-fitted representation geometry to surface or withhold unchanged wearable-stress predictions.
- Evaluated ReliaGate across WESAD, UBFC-Phys, EmpaticaE4Stress, and PhysioNetE4 using subject-disjoint folds, fold-local selection, and paired held-out-subject intervals; on UBFC-Phys, relative to the learned no-geometry router, increased coverage by **10.1 percentage points** while reducing accepted risk by **5.6 percentage points**. Per-subject analyses on WESAD showed that favorable pooled outcomes could coexist with higher median participant-level accepted risk.
- **ReliaGate++ (ongoing follow-on work):** Extending static routing into validation-locked temporal emission control and auditing cue precision, false stress and reassurance, contradiction and flip rates, subject reach, and episode coverage.
- **INTERSPEECH 2026:** Developed a leakage-safe intervention diagnostic that freezes trained experts and fusion parameters while shuffling quality scores across held-out examples to test decision-level reliance. Across StressID and CMU-MOSEI, 25 runs per dataset showed near-zero native performance changes, while positive controls confirmed sensitivity when reliability tracked correctness.

Mayo Clinic Comprehensive Cancer Center

Jun 2026 – Present

Summer Undergraduate Research Fellow | Advisor: Wei Liu, Ph.D.

- Developed a validation-locked **selective-review model** for quality assurance of AI-generated head-and-neck organ contours using CT, geometric, segmentation uncertainty, and native-plan dose-context evidence; classified **59.4% of 3,812 held-out contour instances** as low risk at **99.91% NPV**, **97.47% sensitivity**, and **0.936 AUROC**.
- Built a patient-disjoint training and evaluation pipeline spanning **752 cases**, **37,228 contour instances**, and **5 folds** with nnU-Net and SLURM; added automated checks for split leakage, provenance, and reproducibility.

AIM HIGH Lab, Dartmouth College

Feb 2025 – Aug 2025

Waterhouse Research Award | Advisor: Prof. Nicholas C. Jacobson

- Fine-tuned **LLaMA-3.1-8B-Instruct** with LoRA/QLoRA for therapeutic-dialogue generation and implemented reproducible training and inference workflows under constrained compute.
- Evaluated **1,753 generated responses** using semantic metrics and structured failure analysis to identify contextual grounding and off-target generation errors.

Honors, Awards & Research Funding

Mayo Clinic Summer Undergraduate Research Fellowship (SURF) – \$6,000	2026
Passport AI Grant, GoPassport Education (external support for Mayo Clinic SURF) – \$2,000	2026
Best Class Project Citation – Deep Learning Robustness (COSC 89.31)	Winter 2026
James O. Freedman Presidential Scholar – \$3,400	2025
Waterhouse Research Award (Summer Research Grant) – \$6,700	2025
Dean's Honor List	2023–2026

Technical Skills

Methods: Selective prediction; uncertainty estimation and calibration; distribution-shift evaluation; multimodal fusion; time-series modeling; representation analysis; subject- and patient-disjoint evaluation
ML & Scientific Computing: Python, PyTorch, scikit-learn, Hugging Face Transformers/PEFT, NumPy, pandas, SciPy, nnU-Net
Systems & Reproducibility: Linux, SLURM, Git, AWS, Flask/FastAPI; experiment automation, provenance tracking, and reproducible pipelines
Signals & Data: PPG/BVP, EDA, ECG, temperature; OpenSMILE, NeuroKit2, BioSPPy; DICOM, NIfTI

Selected Technical Project

Local Mental Health Finder Live Demo | Code
• Built and evaluated a full-stack therapist-discovery system using symptom-based search and learning-based provider ranking under sparse, noisy inputs.

Leadership & Service

Republic of Korea Marine Corps (ROKMC) Sep 2021 – Mar 2023
Communications Specialist & Interpreter (Squad Leader) – Hwaseong, South Korea
• Led a **20-person unit** and initiated “Weekly Hour,” a peer-support program designed to identify unspoken distress in a high-stigma environment.
• Managed headquarters networking infrastructure and resolved **150+ IT support cases monthly**; provided technical interpretation during joint U.S.–ROK exercises.

Dartmouth Tech Consulting Group Sep 2024 – Present
• Led web-development and digital-strategy projects for campus organizations, delivering production systems under client and timeline constraints.

Korean Student Association Sep 2023 – Present
• Organized cultural-exchange events and mentorship programs supporting international students.

Languages

Korean (native); English (full professional proficiency)